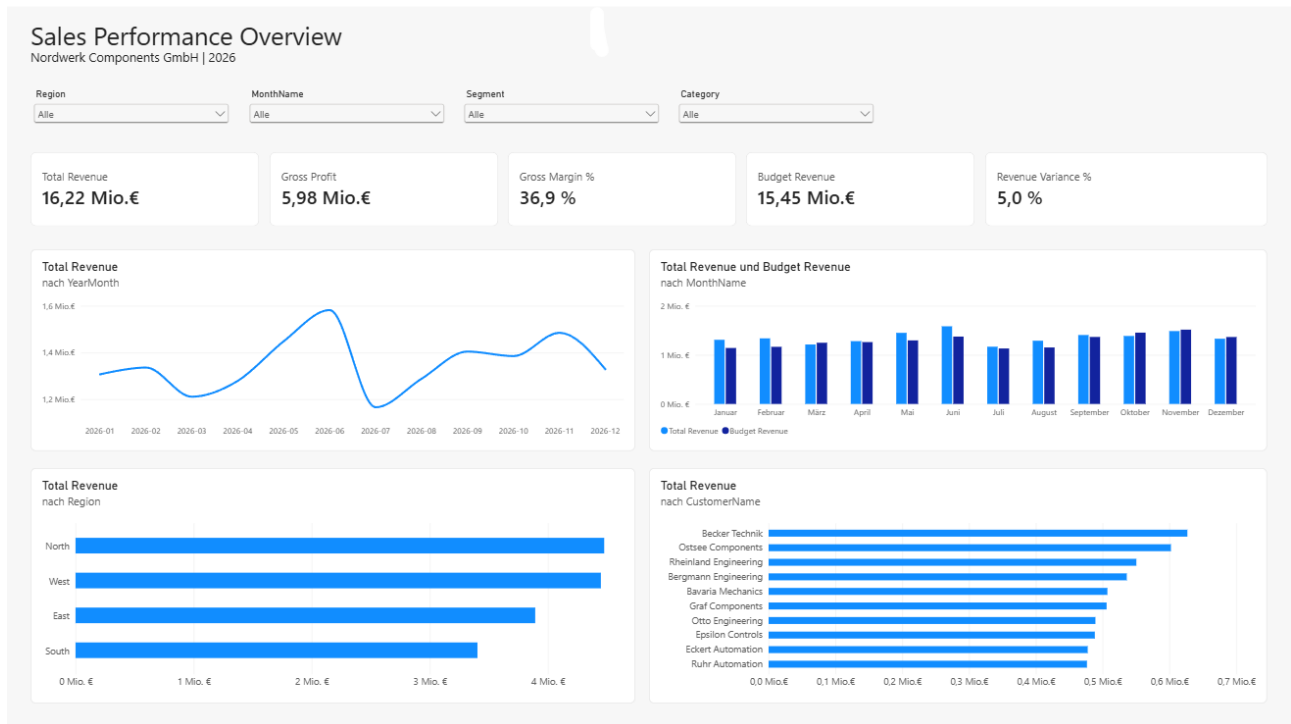


Sales Reporting Automation with Power BI

A portfolio project demonstrating how a manual, Excel-based sales reporting process can be transformed into a repeatable Power BI reporting workflow.



Business Problem

A fictional mid-sized manufacturing company receives monthly sales data as Excel exports.

The previous reporting process requires manual work:

- collecting monthly Excel files
- combining and cleaning data
- enriching sales data with customer and product information
- calculating KPIs
- updating management reports

This makes the reporting process repetitive and increases the risk of inconsistent calculations and manual errors.

Solution

The project transforms the process into a structured reporting workflow:

Excel files

→ Power Query

→ Data Model

→ DAX Measures

→ Power BI Dashboard

The goal is not to introduce unnecessary technology, but to use the appropriate tool for each part of the workflow.

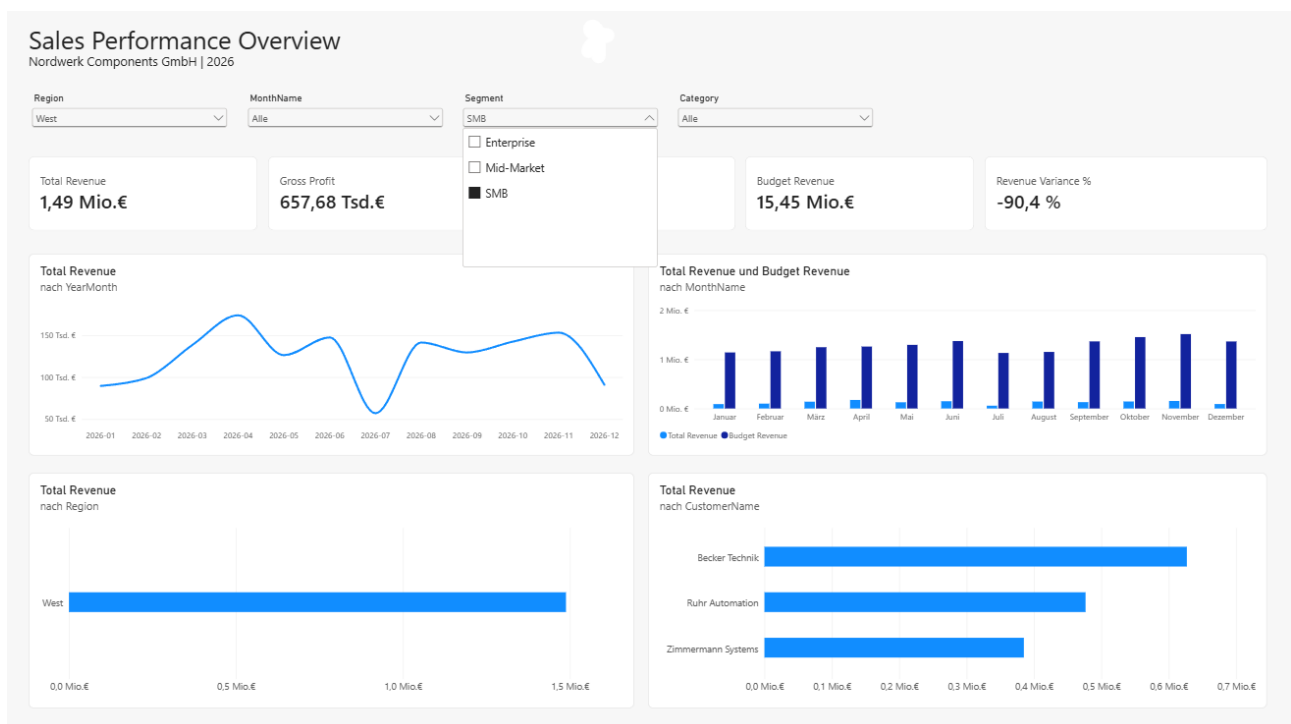
Dashboard

The management dashboard provides an overview of:

- Total Revenue
- Gross Profit
- Gross Margin
- Budget Revenue
- Revenue Variance
- Revenue development over time
- Actual vs. Budget
- Revenue by Region
- Top Customers

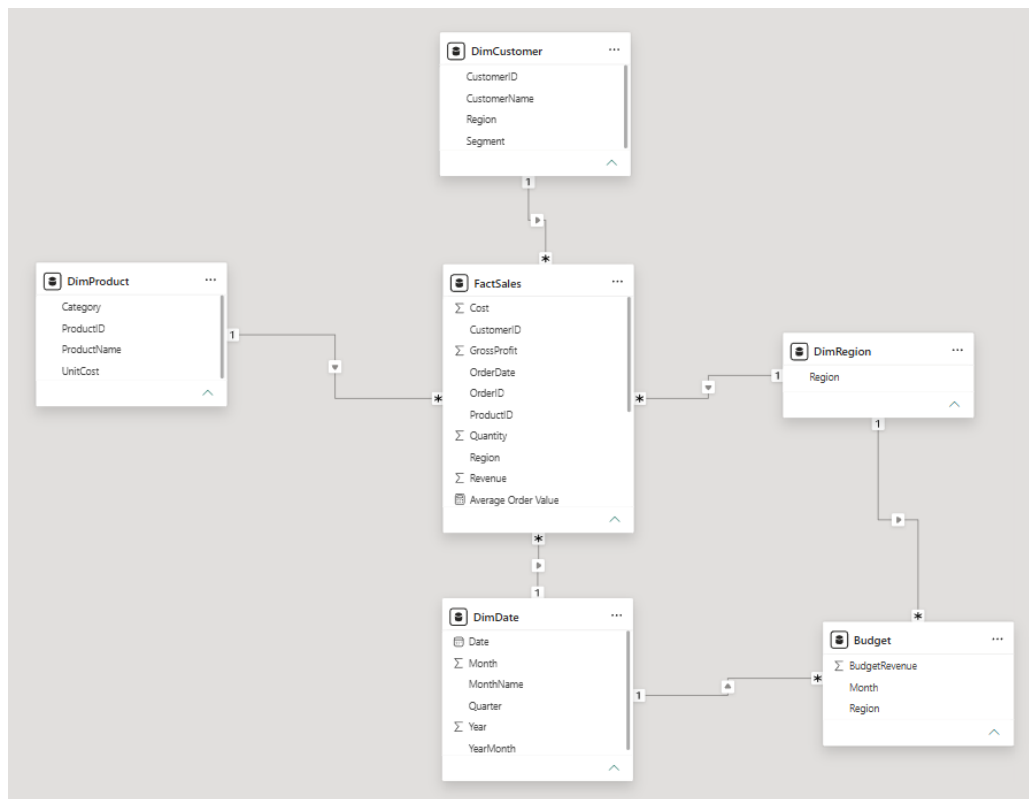
Interactive filters allow the user to analyze the results by:

- Year
- Month
- Region
- Product Category
- Customer Segment



Data Model

The Power BI model follows a simple star-schema approach.



Fact tables

- FactSales
- Budget

Dimension tables

- DimDate
- DimRegion
- DimCustomer
- DimProduct

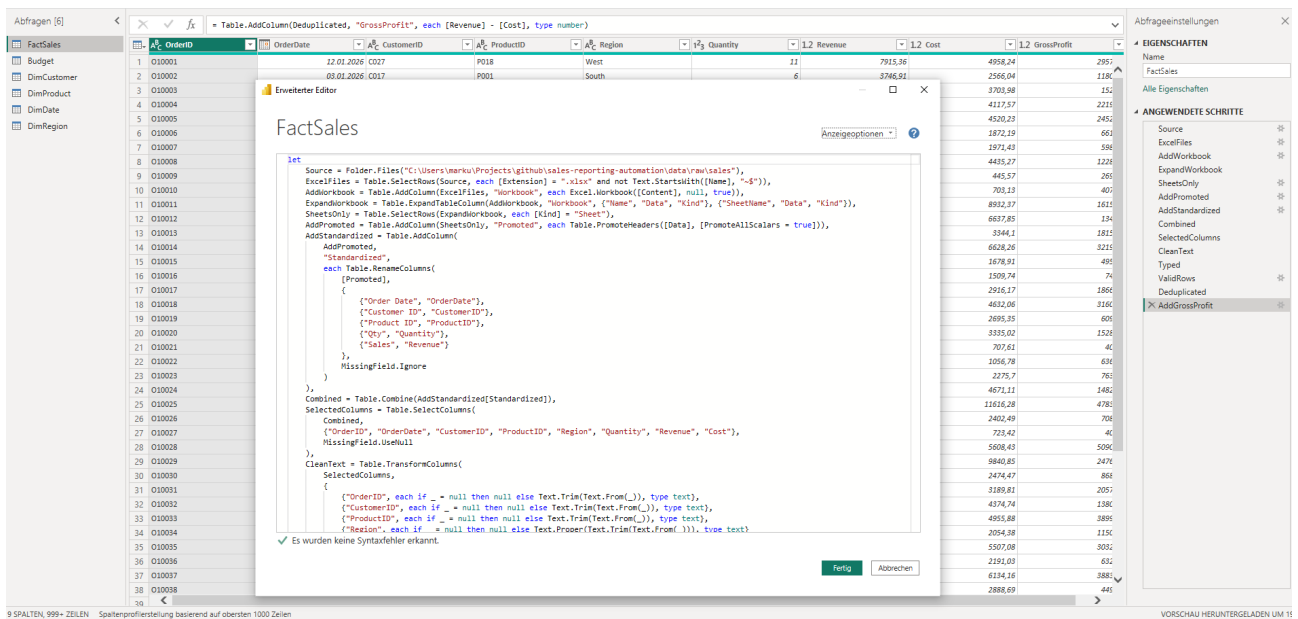
The shared dimensions allow sales and budget data to be analyzed consistently by time, region, customer and product.

Data Transformation

Power Query is used to transform the raw Excel exports into analysis-ready tables.

The transformation layer handles typical issues found in recurring Excel-based reporting processes, such as:

- combining monthly files
- standardizing column names
- cleaning inconsistent values
- handling missing values
- removing duplicate records
- applying consistent data types



Key Metrics

The report uses DAX measures for the main business KPIs, including:

- Total Revenue
- Total Cost
- Gross Profit
- Gross Margin %
- Order Count
- Average Order Value
- Budget Revenue
- Revenue Variance
- Revenue Variance %

Measures are used instead of hard-coded calculations so that the KPIs respond dynamically to report filters.

Business Value

The resulting workflow demonstrates how a repetitive Excel-based reporting process can be turned into a more structured and reusable reporting solution.

Potential benefits include:

- less repetitive manual reporting work
- consistent KPI definitions
- easier Actual-vs-Budget analysis
- interactive management reporting
- easier analysis across regions, customers and products
- a repeatable monthly reporting workflow

This project is a synthetic portfolio demonstration and does not represent a real client implementation. Therefore, no specific time or cost savings are claimed.

Technology

- Power BI
- Power Query / M
- DAX
- Excel
- Git / GitHub

Project Structure

```
sales-reporting-automation/  
|  
├─ data/  
│  └─ raw/  
|  
├─ powerbi/  
│  ├─ SalesReportingAutomation.pbip  
│  ├─ SalesReportingAutomation.Report/  
│  └─ SalesReportingAutomation.SemanticModel/  
|  
├─ screenshots/  
|  
├─ README.md  
└─ .gitignore
```

Project Status

The first management dashboard page is implemented.

Future improvements may include:

- a detailed sales analysis page
- additional management KPIs
- further automation of the data ingestion process
- optional Python/SQL integration where it provides a meaningful business benefit